

Entropic Optimal Transport and Sinkhorn

Exercise statements

This version contains the exercises only. Hints and worked solutions are available in the companion PDF.

Exercise 1 (2x2 entropic OT with perturbation).

Let

$$C = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad a = \begin{pmatrix} p \\ 1-p \end{pmatrix}, \quad b = \begin{pmatrix} \frac{1}{2} \\ \frac{1}{2} \end{pmatrix},$$

with $p \in (0, 1)$, and let $\varepsilon > 0$. Consider

$$\min_{P \in \mathbb{R}_+^{2 \times 2}} \langle C, P \rangle + \varepsilon \sum_{i,j} P_{i,j} (\log P_{i,j} - 1) \quad \text{subject to} \quad P\mathbf{1} = a, \quad P^\top \mathbf{1} = b.$$

(1) Show that every admissible transport plan can be written as

$$P(x) = \begin{pmatrix} x & y(x) \\ z(x) & w(x) \end{pmatrix}$$

for a single scalar x , and give explicit formulas for $y(x)$, $z(x)$, $w(x)$, as well as the admissible interval $I(p)$ for x .

(2) Rewrite the objective as a scalar function $F_\varepsilon(x)$ and compute $F'_\varepsilon(x)$.

(3) Explain why the optimal $x_\varepsilon(p)$ is necessarily in the interior of $I(p)$ for $0 < p < 1$. Optional: from the optimality condition $F'_\varepsilon(x) = 0$, give the expression of the optimal $x_\varepsilon(p)$.

(4) Optional: assume $0 < p < 1/2$. As $\varepsilon \downarrow 0$, obtain an expansion $x_\varepsilon(p) = p + c(p)e^{-2/\varepsilon} + o(e^{-2/\varepsilon})$ and determine $c(p)$.

Exercise 2 (Dual of entropic-regularized linear program).

Consider the entropic-regularized linear program

$$\min_{P \geq 0, A(P)=b} \langle C, P \rangle + \varepsilon H(P), \quad H(P) := \sum_i P_i (\log P_i - 1),$$

where P is a finite-dimensional nonnegative vector (think of $P \in \mathbb{R}_+^N$, or a matrix reshaped as a vector), $C \in \mathbb{R}^N$, $A : \mathbb{R}^N \rightarrow \mathbb{R}^m$ is linear, $b \in \mathbb{R}^m$, and $\varepsilon > 0$. Assume strict feasibility: there exists $P^0 > 0$ with $A(P^0) = b$. Use $0 \log 0 = 0$.

(1) Write the Lagrangian $\mathcal{L}(P, \lambda)$.

(2) Assuming we can swap $\inf_P$ and $\sup_\lambda$, write the dual problem.

(3) Give the primal–dual optimality relation between P and λ .

(4) Give the expression of $A(P)$ in the case of the marginal constraint of classical OT. Compute $A^\top$, and show that the dual you obtained matches the classical dual of entropic OT.

Exercise 3 (Entropic multimarginal OT with 3 marginals).

Let $a \in \mathbb{R}_{++}^{n_1}$, $b \in \mathbb{R}_{++}^{n_2}$, $c \in \mathbb{R}_{++}^{n_3}$ of equal total mass, and let $\varepsilon > 0$ and $C \in \mathbb{R}^{n_1 \times n_2 \times n_3}$. For $T \in \mathbb{R}_+^{n_1 \times n_2 \times n_3}$ define the three marginalization operators

$$\pi_1(T) := \left(\sum_{j,k} T_{i,j,k} \right)_{i=1}^{n_1}, \quad \pi_2(T) := \left(\sum_{i,k} T_{i,j,k} \right)_{j=1}^{n_2}, \quad \pi_3(T) := \left(\sum_{i,j} T_{i,j,k} \right)_{k=1}^{n_3}.$$

Consider the problem

$$\min_{T \geq 0} \langle C, T \rangle + \varepsilon \sum_{i,j,k} T_{i,j,k} (\log T_{i,j,k} - 1) \quad \text{s.t.} \quad \pi_1(T) = a, \pi_2(T) = b, \pi_3(T) = c.$$

- (1) Write the Lagrangian using multipliers f, g, h for π_1, π_2, π_3 .
- (2) Assuming one can swap $\min_T$ and $\max_{f,g,h}$, write the dual.
- (3) Write the Sinkhorn iterations in the log domain for the three potentials.
- (4) (Bonus) Explain why the usual Hilbert-metric/Birkhoff argument for 2 marginals does not give a contraction here.

Exercise 4 (Gaussian example for Sinkhorn in dual form).

Let α and β be probability measures on $\mathbb{R}$ and let $c(x, y) = \frac{1}{2}(x - y)^2$. The entropic OT dual can be written

$$\max_{f,g} \int f(x) d\alpha(x) + \int g(y) d\beta(y) - \varepsilon \int \int \exp\left(\frac{f(x)+g(y)-c(x,y)}{\varepsilon}\right) d\alpha(x) d\beta(y).$$

- (1) Recall the Sinkhorn update on f (dual form).
- (2) Assume $\alpha = \mathcal{N}(m_1, \sigma_1^2)$, $\beta = \mathcal{N}(m_2, \sigma_2^2)$, with $\sigma_1, \sigma_2 > 0$, and let $g(y) = \frac{1}{2}ay^2 + by + c_0$, where $a < 1 + \varepsilon/\sigma_2^2$ (the integrability condition). Compute the integral on the right-hand side and show that the updated f is also quadratic.
- (3) Give an intuitive explanation of why (for this quadratic/Gaussian setting) the optimal dual potentials remain quadratic, and what this implies for the optimal coupling $\pi^\star$.

Exercise 5 (Envelope theorem and gradients of the entropic cost).

Recall the envelope theorem in the finite-dimensional smooth case:

Let $V(\theta) := \min_z F(z, \theta)$ with F smooth, with the minimization over a fixed compact set, and for each θ assume there is a unique minimizer $z^\star(\theta)$. Then

$$\nabla_\theta V(\theta) = \nabla_\theta F(z^\star(\theta), \theta),$$

i.e. the derivative of V w.r.t. θ is the partial derivative of F at the optimizer (no derivative of $z^\star$ needed).

Let $a \in \mathbb{R}_{++}^m$, $b \in \mathbb{R}_{++}^n$ be probability histograms, let $\varepsilon > 0$, and $C \in \mathbb{R}^{m \times n}$ a cost matrix. The entropic OT cost is

$$\mathcal{W}_\varepsilon(a, b; C) := \min_{\substack{P \in \mathbb{R}_+^{m \times n} \\ P\mathbf{1} = a, P^\top \mathbf{1} = b}} \left\{ \langle C, P \rangle + \varepsilon \sum_{i,j} P_{i,j} (\log P_{i,j} - 1) \right\}.$$

- (1) Using the envelope theorem on this *primal* problem, give the gradient $\nabla_C \mathcal{W}_\varepsilon(a, b; C)$.
- (2) Assume the cost comes from positions: $x = (x_1, \dots, x_m) \subset \mathbb{R}^d$, $y = (y_1, \dots, y_n) \subset \mathbb{R}^d$, and

$$C_{i,j} = \frac{1}{2} \|x_i - y_j\|^2.$$

Compute $\nabla_{x_1} \mathcal{W}_\varepsilon(a, b; C)$.

(3) Recall the dual of entropic OT

$$\mathcal{W}_\varepsilon(a, b; C) = \max_{f \in \mathbb{R}^m, g \in \mathbb{R}^n} \langle f, a \rangle + \langle g, b \rangle - \varepsilon \sum_{i,j} \exp\left(\frac{f_i + g_j - C_{i,j}}{\varepsilon}\right).$$

Using again the envelope theorem, compute its derivative along perturbations $\dot{a}$ satisfying $\sum_i \dot{a}_i = 0$, and explain the additive-constant ambiguity.

Exercise 6 (Entropic OT, joint convexity, and semi-relaxed reformulation).

Let $a = (a_1, \dots, a_m) \in \Delta_m$ and $b = (b_1, \dots, b_n) \in \Delta_n$ be probability vectors and let $C \in \mathbb{R}^{m \times n}$ be a cost matrix. Consider the (balanced) entropic OT problem

$$W_\varepsilon(a, b) := \min_{\substack{P \in \mathbb{R}_+^{m \times n} \\ P \mathbf{1}_m = a, P^\top \mathbf{1}_n = b}} \left\{ \langle C, P \rangle + \varepsilon H(P) \right\}, \quad H(P) := \sum_{i,j} P_{i,j} (\log P_{i,j} - 1).$$

It is well known (and you may take it for granted) that the dual of this problem is

$$W_\varepsilon(a, b) = \sup_{f \in \mathbb{R}^m, g \in \mathbb{R}^n} \left\{ \langle f, a \rangle + \langle g, b \rangle - \varepsilon \sum_{i=1}^m \sum_{j=1}^n \exp\left(\frac{f_i + g_j - C_{i,j}}{\varepsilon}\right) \right\}.$$

(1) Using the dual expression above, show that *both* maps

$$a \mapsto W_\varepsilon(a, b) \quad \text{and} \quad b \mapsto W_\varepsilon(a, b)$$

are convex, and in fact that $W_\varepsilon(a, b)$ is jointly convex in (a, b) .

(2) Show that the outer problem

$$\min_{a \in \Delta_m} W_\varepsilon(a, b)$$

is equivalent to the semi-relaxed problem

$$\min_{\substack{P \geq 0 \\ P^\top \mathbf{1}_m = b}} \langle C, P \rangle + \varepsilon H(P).$$

(3) Solve this semi-relaxed problem in closed form.

Exercise 7 (Invariance and rescaling (continuous setting)).

Let X, Y be compact metric spaces, α, β probability measures, and $c : X \times Y \rightarrow \mathbb{R}$ a continuous cost and $\varepsilon > 0$, consider the entropic optimal transport problem

$$\min_{\pi \in \Pi(\alpha, \beta)} \left\{ \int_{X \times Y} c(x, y) d\pi(x, y) + \varepsilon \int_{X \times Y} \left(\log \frac{d\pi}{d(\alpha \otimes \beta)}(x, y) - 1 \right) d\pi(x, y) \right\},$$

where $\Pi(\alpha, \beta)$ is the set of couplings of α and β and the entropy is defined as $+\infty$ if $\pi \not\ll \alpha \otimes \beta$.

(1) **(Invariance under potentials)** Let $u : X \rightarrow \mathbb{R}$ and $v : Y \rightarrow \mathbb{R}$ be integrable. Show that replacing the cost by

$$c'(x, y) = c(x, y) + u(x) + v(y)$$

does *not* change the optimal plan: the minimizer for c' is the same coupling as for c .

(2) **(Scaling of the cost)** For $\lambda > 0$, denote by $\pi_{c, \varepsilon}$ the optimal coupling for cost c and regularization ε , and by $\pi_{\lambda c, \varepsilon}$ the one for the scaled cost λc . Show that

$$\pi_{\lambda c, \varepsilon} = \pi_{c, \varepsilon/\lambda},$$

i.e. *scaling the cost by λ is the same as keeping the cost and scaling the regularization by $1/\lambda$.*

Further exercises

Exercise 8 (Unbalanced OT as constrained entropic problem).

Let $C \in \mathbb{R}^{m \times n}$, $a \in \mathbb{R}_+^m$, $b \in \mathbb{R}_+^n$, $\tau > 0$. Define, for $r \in \mathbb{R}_+^m$, $a \in \mathbb{R}_+^m$,

$$\text{KL}(r \mid a) := \sum_{i=1}^m \left(r_i \log \frac{r_i}{a_i} - r_i + a_i \right),$$

and similarly for $\text{KL}(s \mid b)$. Consider the unbalanced OT problem

$$\min_{P \in \mathbb{R}_+^{m \times n}} \langle C, P \rangle + \tau \text{KL}(P \mathbf{1}_n \mid a) + \tau \text{KL}(P^\top \mathbf{1}_m \mid b). \quad (\text{UOT})$$

(1) Show that (UOT) is equivalent to

$$\min_{\substack{P \geq 0 \\ P \mathbf{1}_n = r \\ P^\top \mathbf{1}_m = s}} \langle C, P \rangle + \tau \text{KL}(r \mid a) + \tau \text{KL}(s \mid b), \quad (\star)$$

and argue that if $(P^\star, r^\star, s^\star)$ solves $(\star)$, then $P^\star$ is an optimal *balanced* OT plan between the (relaxed) marginals $(r^\star, s^\star)$.

- (2) Write the Lagrangian of $(\star)$ using dual vectors $f \in \mathbb{R}^m$, $g \in \mathbb{R}^n$ for the constraints $P \mathbf{1}_n = r$ and $P^\top \mathbf{1}_m = s$.
- (3) Assuming we can swap $\inf_{P, r, s}$ and $\sup_{f, g}$, derive the dual involving only the two potentials f, g .
- (4) Assume a, b have equal total mass. Show that the dual objective tends pointwise to the balanced OT dual objective as $\tau \rightarrow +\infty$, and justify convergence of the optimal values.

Exercise 9 (Unbalanced OT + entropic regularization).

Let $C \in \mathbb{R}^{m \times n}$, $a \in \mathbb{R}_+^m$, $b \in \mathbb{R}_+^n$, and let $\tau > 0$, $\varepsilon > 0$. Denote

$$H(P) := \sum_{i,j} P_{i,j} (\log P_{i,j} - 1), \quad \text{KL}(r \mid a) := \sum_i \left(r_i \log \frac{r_i}{a_i} - r_i + a_i \right).$$

Consider

$$\min_{P \geq 0} \langle C, P \rangle + \varepsilon H(P) + \tau \text{KL}(P \mathbf{1} \mid a) + \tau \text{KL}(P^\top \mathbf{1} \mid b).$$

- (1) Compute the dual problem and show it involves only two dual potentials $f \in \mathbb{R}^m$, $g \in \mathbb{R}^n$.
- (2) For fixed g , give the maximizer in f (unbalanced Sinkhorn update).
- (3) Assume equal input masses and identify the pointwise limit of the dual objective as $\tau \rightarrow +\infty$.

Exercise 10 (φ -divergence, dual form and unbalanced OT).

Let α and β be finite nonnegative measures on a measurable space X , and let $\varphi : \mathbb{R}_+ \rightarrow \mathbb{R} \cup \{+\infty\}$ be finite, convex and lower semicontinuous on $[0, \infty)$, with $\varphi(1) = 0$ and $\lim_{r \rightarrow \infty} \varphi(r)/r = +\infty$. The φ -divergence of α w.r.t. β is

$$D_\varphi(\alpha \mid \beta) := \begin{cases} \int \varphi\left(\frac{d\alpha}{d\beta}(x)\right) d\beta(x), & \alpha \ll \beta, \\ +\infty, & \text{otherwise.} \end{cases}$$

- (1) Recall the convex conjugate $\varphi^\star(t) := \sup_{u \geq 0} (ut - \varphi(u))$.

(2) Show that

$$D_\varphi(\alpha \mid \beta) = \sup_f \left\{ \int f d\alpha - \int \varphi^\star(f) d\beta \right\}.$$

(3) Consider now the unbalanced OT problem

$$\min_{\pi \geq 0} \int c(x, y) d\pi(x, y) + D_\varphi(\pi_1 \mid \alpha) + D_\varphi(\pi_2 \mid \beta),$$

where π_1, π_2 are the first and second marginals of π . Using (2), derive a dual lower bound involving only two measurable test functions f, g . Show that equality holds whenever the indicated minimax interchange is justified; do not assert strong duality on an arbitrary measurable space without further hypotheses.

Exercise 11 (Barycentric projection from entropic OT).

Let $\alpha = \sum_{i=1}^m a_i \delta_{x_i}$ and $\beta = \sum_{j=1}^n b_j \delta_{y_j}$ be probability measures with distinct support points and strictly positive weights, and let P_ε be the optimal plan of entropic OT between the weight vectors (a, b) for some finite cost $C_{i,j} = c(x_i, y_j)$. For x_i with $a_i > 0$ define the barycentric projection

$$T_\varepsilon(x_i) := \frac{1}{a_i} \sum_{j=1}^n (P_\varepsilon)_{i,j} y_j.$$

(1) Suppose the optimal unregularized coupling is uniquely induced by a map $T : \{x_i\} \rightarrow \{y_j\}$. Denote the matrix of its graph plan $(\text{Id}, T)_\# \alpha$ by $P^\star$, and assume $P_\varepsilon \rightarrow P^\star$ entrywise as $\varepsilon \downarrow 0$. Show that

$$T_\varepsilon(x_i) \longrightarrow T(x_i) \quad \text{for every } i.$$

(2) Assume now that $c(x, y) = \frac{1}{2} \|x - y\|^2$ and recall from Exercise 5 that

$$\nabla_{x_i} \mathcal{W}_\varepsilon(a, b) = a_i x_i - \sum_{j=1}^n (P_\varepsilon)_{i,j} y_j.$$

Express $T_\varepsilon(x_i)$ in terms of $\nabla_{x_i} \mathcal{W}_\varepsilon(a, b)$ and x_i , and explain why the displacement $T_\varepsilon(x_i) - x_i$, rather than the target point itself, is the negative cost gradient divided by a_i .